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Department of Computer Science

Academic Year: 2025-26

S.Y.B.Sc.(Data Science) Sem-IV

Subject: DS-252-MJ-T : Data Structure-II

Theory Assignment - 1

Date: 11/12/2025

Attempt all of the following:

1. **Define:** Ancestor, Descendent, Sibling, Leaf node/External node, Non leaf node or Internal node, Binary tree, Complete binary tree, Skewed Binary tree, Left skewed binary tree, Right skewed binary tree, Perfect binary tree, Expression binary tree, Degree of a node, Degree of a tree,
2. The indegree of the root node of a tree is always zero. Justify (T/F).
3. Balance factor every leaf node is zero. Justify (T/F).
4. Explain node structure of binary tree.
5. What is binary search tree?
6. Explain tree traversal techniques in detail.
7. Write applications of a tree.
8. What do you mean by Huffman Encoding?
9. Construct binary search tree for: 13, 4, 25, 3, 21, 20, 7
10. Write the steps for creating BT for the following data: 22, 13, 4, 6, 25, 23, 20, 18, 7, 27
11. Write the steps for creating BST for the following data: 22, 13, 4, 6, 25, 23, 20, 18, 7, 27
12. Define min heap and max heap. Construct heap tree for : 32, 15, 64, 2, 75, 67, 57, 80
13. Traverse the following binary tree using given traversal technique: Inorder, Preorder and Postorder

